

AYUSH SHAH

Software Engineer | AI Agents & Systems Developer

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Summary

Systems & AI Software Engineer specialized in building terminal-native AI coding agents, high-performance backends (Rust, Go), and ROS 2 robotics control pipelines. Experienced in engineering low-latency network protocols, cloud automation tooling, and real-time web architectures.

Experience

Robotics / Systems Software Intern

2025 – Present

Roboparadigm, Hyderabad, India

- Developed and simulated advanced motion planning algorithms for 7DOF and OpenManipulator-X robotic arms using **ROS 2**, **MovelT Task Constructor (MTC)**, and **Gazebo**.
- Optimized kinematics, collision avoidance, and trajectory execution pipelines, reducing path generation latency for automated pick-and-place tasks.
- Integrated real-time sensor feedback loops with ROS 2 nodes for precision spatial manipulation.

Education

B.Tech in Computer Science and Engineering

Expected 2028

Keshav Memorial College of Engineering, Hyderabad, Telangana

Technical Skills

Languages	TypeScript, Rust, Go, C/C++, Python, JavaScript, Bash, Lua
AI & Robotics	AI Coding Agents, LLM Integration, ROS 2, MovelT / MTC, Gazebo
Systems & Web	React, Next.js, Tokio, Express, Actix Web, WebSockets, REST APIs
Databases & DevOps	PostgreSQL, Redis, Docker, Git, Cloudflare Workers, AWS, Nginx, Linux

Selected High-Impact Projects

Forge | Terminal-Native AI Coding Agent

🐙 GitHub 🌐 Live

Tech Stack: TypeScript, Node.js, LLMs, CLI Architecture, TailwindCSS

- Architected an autonomous CLI AI agent capable of multi-file code editing, AST context building, shell execution, and pair programming directly inside the terminal.
- Engineered context-aware file indexers and error trace parsers to enable accurate self-healing code generation.

7DOF MTC Arm Control | Robotic Motion Planning

🐙 GitHub

Tech Stack: C++, ROS 2, MovelT Task Constructor, Gazebo, Docker

- Designed a MovelT Task Constructor (MTC) package for 7-degree-of-freedom manipulators, enabling multi-stage trajectory planning and obstacle-aware manipulation.
- Containerized ROS 2 workspace environments in Docker for reproducible robotic simulations.

Cloud-It | Cloud Infrastructure Automation Utility

🐙 GitHub

Tech Stack: TypeScript, Node.js, AWS (EC2, S3), Docker, DevOps

- Engineered a TypeScript infrastructure automation tool to streamline cloud workload deployment, container orchestration, and environment configuration across AWS services.

rust-smtp & TempMail | Asynchronous Mail Engine

🐙 GitHub 🌐 Live

Tech Stack: Rust, Tokio, Bun, Express, PostgreSQL, React, Nginx

- Built a custom high-throughput SMTP server in Rust using Tokio for asynchronous, non-blocking email processing and MIME parsing.
- Powered a disposable email platform with real-time inbox streaming via PostgreSQL & Bun backend.

Raven | Real-Time High-Concurrency Chat Platform

🐙 GitHub 🌐 Live

Tech Stack: Next.js, TypeScript, WebSockets, Redis, PostgreSQL, Prisma, Turborepo

- Engineered a scalable real-time messaging platform combining WebSockets for instant data propagation and REST APIs for authentication and fallback.

shrty | Low-Latency Link Shortener Microservice

 GitHub  Live

Tech Stack: Go, PostgreSQL, React, Cloudflare Turnstile, Vite

- Developed a microsecond-latency URL shortening microservice using Go, with Cloudflare Turnstile verification against bot abuse.